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Drug-Induced Interstitial Lung Disease Associated with Biological and Targeted DMARDs: A Two-Case Clinical Observation

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ABSTRACT

Biological disease-modifying antirheumatic drugs (DMARDs) targeting TNF-alpha and small-molecule inhibitors have revolutionized management of chronic inflammatory arthritides. However, pulmonary parenchymal toxicity—including drug-induced interstitial lung disease (DIILD) and cryptogenic organizing pneumonia—represents a severe diagnostic challenge. In this report, we evaluate non-case aggregate registry cohorts and document two distinct individual patient cases of life-threatening pulmonary toxicity occurring secondary to Infliximab and Leflunomide, highlighting the imperative for individualized safety disaggregation.

1. INTRODUCTION & REGISTRY SURVEILLANCE

Pulmonary involvement in rheumatic diseases can stem from primary underlying autoimmune disease or secondary pharmacotherapy toxicity. Disentangling underlying disease progression from iatrogenic pulmonary injury requires meticulous clinicoradiological correlation.

2. OBSERVATIONAL REGISTRY ANALYSIS (NON-CASE COHORT)

We initially reviewed surveillance telemetry from the Mid-Atlantic Rheumatic Disease Registry comprising 420 aggregate patients receiving biologic or targeted synthetic DMARDs over a 5-year observation interval. In this aggregate cohort, non-specific respiratory complaints (mild dry cough, dyspnea on exertion) occurred in 14.2% of patients without verifiable objective parenchymal infiltrates. These aggregate registry statistics do not represent individual ICSR safety events.

3. PATIENT CASE REPORT 1: INFLIXIMAB PNEUMONITIS

A 54-year-old male (Patient T.K., weight 75 kg) with a 6-year history of seropositive rheumatoid arthritis was initiated on Infliximab (Remicade) 5 mg/kg IV infusions at weeks 0, 2, and 6, followed by 8-week maintenance.

Twelve days following his fourth infusion, the patient presented with acute high fever (38.9 deg C), rapidly progressive hypoxemic respiratory failure (PaO2 54 mmHg on room air), and severe non-productive cough. High-resolution chest CT (HRCT) demonstrated bilateral extensive ground-glass opacities with subpleural sparing. Bronchoalveolar lavage (BAL) ruled out Pneumocystis jirovecii, fungal pathogens, and viral etiologies, showing marked lymphocytic alveolitis (CD4:CD8 ratio 3.8).

The patient was admitted to the Medical Intensive Care Unit (MICU) and required non-invasive positive pressure ventilation for 48 hours. Infliximab was permanently withdrawn. High-dose systemic methylprednisolone (125 mg IV every 6 hours) was administered, resulting in progressive radiological and clinical improvement over 14 days. Final Diagnosis: Acute drug-induced interstitial pneumonitis. Seriousness: Hospitalization: YES, Life-Threatening: YES.

4. PATIENT CASE REPORT 2: LEFLUNOMIDE-INDUCED COP

A 41-year-old female (Patient M.S., weight 58 kg) with active psoriatic arthritis was prescribed Leflunomide (Arava) 20 mg PO once daily. Prior pulmonary history was completely unremarkable.

At 12 weeks of continuous therapy, the patient developed subacute progressive exertional dyspnea, low-grade malaise, and bilateral pleuritic chest discomfort. Pulmonary function testing demonstrated restrictive physiology with a 28% decline in DLCO. Chest HRCT revealed patchy peripheral and peribronchovascular consolidations consistent with Cryptogenic Organizing Pneumonia (COP / BOOP).

Leflunomide was immediately ceased. Due to its long terminal half-life (14–16 days), an accelerated elimination washout protocol with oral Cholestyramine (8 g three times daily for 11 days) was executed. Prednisone 40 mg PO QD was co-administered. Follow-up imaging at 8 weeks confirmed complete clearing of consolidations. Seriousness: Inpatient Hospitalization: YES. Positive dechallenge was documented.

5. COMPARATIVE CLINICAL & DIAGNOSTIC MATRIX

Table 1 outlines the comparative clinical, radiological, and therapeutic parameters of the two reported cases:

Case 1 (Infliximab): Male, 54y, 75kg | Onset: 12 days post-infusion | HRCT: Diffuse bilateral ground-glass | Management: MICU, Pulse IV steroids | Outcome: Resolved.

Case 2 (Leflunomide): Female, 41y, 58kg | Onset: 12 weeks | HRCT: Patchy peribronchovascular consolidations (COP) | Management: Cholestyramine washout + oral steroids | Outcome: Resolved.

6. PHARMACOVIGILANCE & REGULATORY IMPLICATIONS

This clinical series emphasizes two fundamental principles for automated safety intake systems:

First, screening systems must effectively filter out non-case background literature sections (such as the 420-patient observational cohort described in Section 2) that do not describe identifiable patient cases.

Second, multicase articles must be parsed into separate, independent Individual Case Safety Reports (ICSRs). Case 1 and Case 2 involve different chemical entities, distinct mechanisms of lung injury, differing onset timelines, and divergent rescue protocols (pulse steroids vs. cholestyramine washout). Aggregating them into a single record violates regulatory compliance.

7. CONCLUSION & SURVEILLANCE RECOMMENDATIONS

Clinicians initiating biological and targeted DMARDs must maintain a high index of suspicion for parenchymal drug toxicities. Automated pharmacovigilance intake systems must incorporate advanced NLP and layout analysis to distinguish between aggregate registry summaries and actionable individual patient cases.

8. REFERENCES

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